ANTU LSS PRO

COMPLETE USER GUIDE

Institutional Liquidity Sweep & Fair Value Gap Execution System | Pine Script v6

Version 1.0 | TradingView Pine Script v6

Table of Contents

1. Introduction
2. What is ANTU LSS PRO?
3. What is Liquidity?
4. Sell-Side Liquidity (SSL)
5. Buy-Side Liquidity (BSL)
6. Liquidity Sweep Mechanics
7. Fair Value Gap (FVG) Mechanics
8. ANTU LSS PRO System Workflow
9. The LONG Setup (Step-by-Step)
10. The SHORT Setup (Step-by-Step)
11. Entry Logic & Consequent Encroachment
12. Stop Loss Calculation (ATR Buffer)
13. Take Profit Calculation (Fixed R:R Multiplier)
14. Risk-to-Reward Ratio (R:R) Dynamics
15. Average True Range (ATR) Volatility Integration
16. Session Settings & Market Time Filters
17. Indicator Settings & Input Parameters
18. How to Read the Chart (Visual Elements & Table)
19. Common Trading Mistakes to Avoid
20. Professional Risk Management Rules
21. Native TradingView Alerts Integration
22. Frequently Asked Questions (FAQ)
23. Official Risk Disclaimer

1. Introduction

Welcome to the official User Guide for **ANTU LSS PRO**.

Modern electronic financial markets are not driven by arbitrary squiggly lines or delayed mathematical oscillators; they are driven by the search for liquidity and price efficiency. Market makers, institutional algorithms, and large liquidity providers operate where order volume is dense. Retail stop-loss clusters resting above recent highs and below recent lows represent prime targets for liquidity accumulation.

ANTU LSS PRO was engineered to translate core institutional price action concepts—specifically **Liquidity Sweeps** and **Fair Value Gaps (FVG)**—into a systematic, objective, non-repainting TradingView indicator.

This guide provides an exhaustive technical breakdown of every component, input setting, formula, and visual indicator element programmed inside the script.

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2. What is ANTU LSS PRO?

ANTU LSS PRO is a standalone, client-side Pine Script v6 indicator written specifically for TradingView. It operates directly inside your web browser, desktop app, or mobile app without requiring external servers, third-party software, or broker API bridges.

Core Capabilities:

- **Automatic Swing Point Detection:** Continuously detects swing highs and swing lows using user-defined pivot lengths.
- **Dynamic Liquidity Tracking:** Stores active, un-swept Buy-Side Liquidity (BSL) and Sell-Side Liquidity (SSL) levels in internal memory arrays.
- **Wick Rejection Verification:** Confirms genuine liquidity sweeps when price breaches a level during the candle but closes back within the structural range.
- **Imbalance Scanning:** Identifies 3-bar Fair Value Gaps that develop in the direction of the post-sweep reversal.
- **50% Retest Execution:** Automatically plots trade setups when price retraces to the exact midpoint (Consequent Encroachment) of the qualifying FVG.
- **Dynamic Risk Boxes:** Visually renders green Take-Profit target boxes, red Stop-Loss boxes, and dashed entry lines.
- **Single-Position Discipline (`hasOpen`):** Suppresses overlapping or duplicate trade signals while a position remains active.
- **Live On-Chart Statistics:** Maintains a rolling performance dashboard reporting current market bias, win rate %, wins, losses, and open setups.
- **Real-Time Alert Dispatch:** Dispatches instant notifications via native TradingView popups and structured JSON alert payloads.

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3. What is Liquidity?

In technical trading, **liquidity** refers to the concentration of pending orders resting in the order book at specific price levels.

Because large institutional market participants trade massive position sizes, they cannot simply enter the market with market orders without causing extreme slippage. To accumulate a large long position, they require an equal volume of sell orders; to distribute a large short position, they require an equal volume of buy orders.

Where do these opposing orders cluster?

1. Directly above prominent swing highs (where breakout buyers buy and short sellers place stop-losses).
2. Directly below prominent swing lows (where breakdown sellers short and long buyers place stop-losses).

These clusters are referred to as **Buy-Side Liquidity (BSL)** and **Sell-Side Liquidity (SSL)**.

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4. Sell-Side Liquidity (SSL)

Sell-Side Liquidity consists of accumulated sell stop orders located beneath swing lows.

How ANTU LSS PRO Tracks SSL:

1. The script identifies a structural pivot low using `ta.pivotlow(low, swingLen, swingLen)`.
2. When a pivot low is confirmed, its price is pushed into the `sslLvls` array.
3. The level remains active (`sslSwept = false`) until price sweeps through it.

4. ANTU LSS PRO tracks up to `maxLevels` active SSL levels simultaneously, discarding older levels via FIFO array trimming.

When retail traders buy an asset, they typically place their protective stop-loss just under the recent swing low. A stop-loss on a buy position is an order to **sell at market**. When price falls below that swing low, thousands of market sell orders are triggered, creating a dense pool of Sell-Side Liquidity.

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5. Buy-Side Liquidity (BSL)

Buy-Side Liquidity consists of accumulated buy stop orders located above swing highs.

How ANTU LSS PRO Tracks BSL:

1. The script identifies a structural pivot high using `ta.pivothigh(high, swingLen, swingLen)`.
2. When a pivot high is confirmed, its price is pushed into the `bslLvls` array.
3. The level remains active (`bslSwept = false`) until price sweeps through it.
4. ANTU LSS PRO tracks up to `maxLevels` active BSL levels in memory.

Traders holding short positions place protective stops above swing highs. A stop-loss on a short position is an order to **buy at market**. Furthermore, breakout traders place buy-stop entry orders above the same high. When price surges above the high, this massive liquidity pool is triggered.

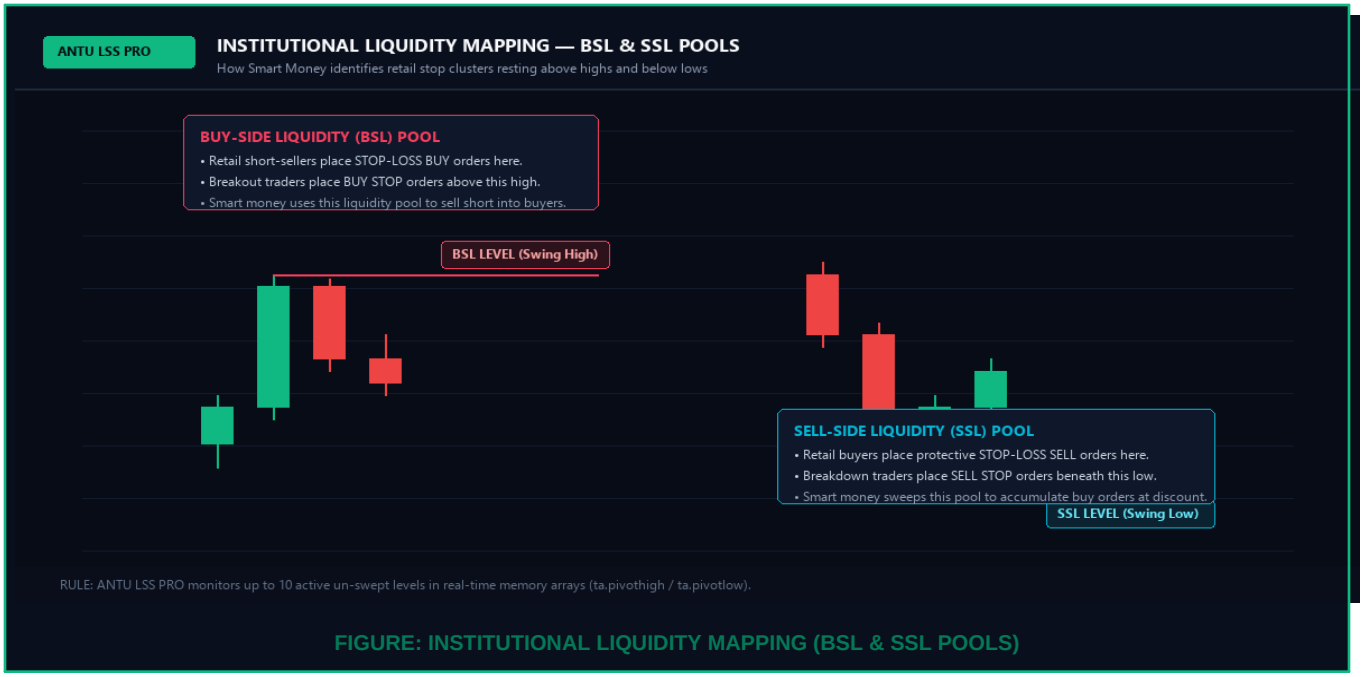


FIGURE: INSTITUTIONAL LIQUIDITY MAPPING (BSL & SSL POOLS)

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6. Liquidity Sweep Mechanics

A **Liquidity Sweep** (also known as a turtle soup or stop run) occurs when price temporarily penetrates a liquidity pool to absorb pending orders, but immediately fails to sustain momentum and closes back on the opposite side of the level.

Mathematical Confirmation in ANTU LSS PRO:

- **SSL Sweep (Bullish Rejection):**

low < Level AND close > Level

Price probed below the pivot low during the candle, triggered the sell stops, but institutional absorption pushed the closing price back above the level. The level is marked as swept, and the candle's lowest price is recorded as **`sweepLow`**.

- **BSL Sweep (Bearish Rejection):**

high > Level AND close < Level

Price spiked above the pivot high during the candle, triggered the buy stops, but institutional distribution forced the closing price back below the level. The level is marked as swept, and the candle's highest price is recorded as **`sweepHigh`**.

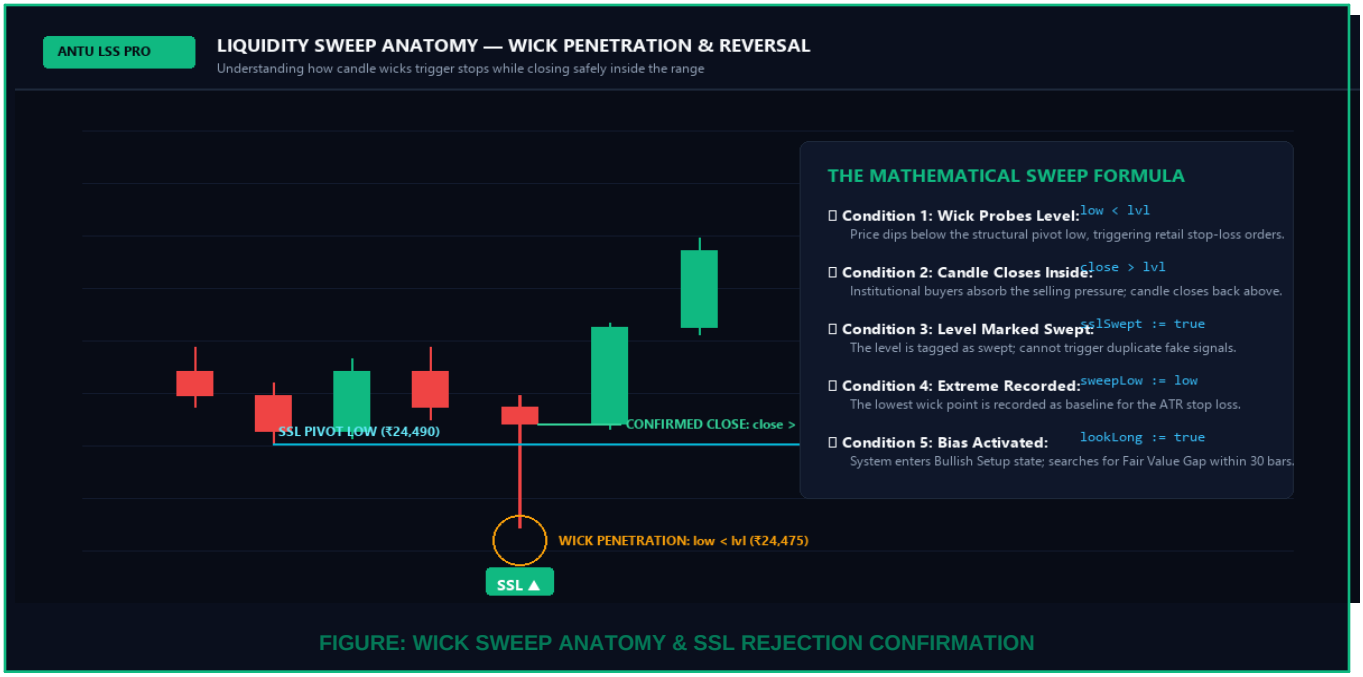


FIGURE: WICK SWEEP ANATOMY & SSL REJECTION CONFIRMATION

7. Fair Value Gap (FVG) Mechanics

A **Fair Value Gap** represents a 3-candle price imbalance caused by rapid, one-sided market orders. When institutional volume enters the market aggressively, price displaces rapidly, leaving an imbalance where buyers and sellers were not matched evenly.

Algorithmic Definition:

- **Bullish FVG (Under-priced Imbalance):**

low[0] > high[2]

The low of candle 3 is strictly higher than the high of candle 1. The gap between **`high[2]`** (bottom) and **`low[0]`** (top) represents un-traded price territory.

- **Bearish FVG (Over-priced Imbalance):**

high[0] < low[2]

The high of candle 3 is strictly lower than the low of candle 1. The gap between **`high[0]`** (bottom) and **`low[2]`** (top) represents unfilled selling imbalance.

ANTU LSS PRO records up to 20 recent bullish and bearish FVGs in rolling arrays (**`bFvgTop`**, **`bFvgBot`**, **`bFvgBar`**, etc.), retaining only recent structural imbalances.

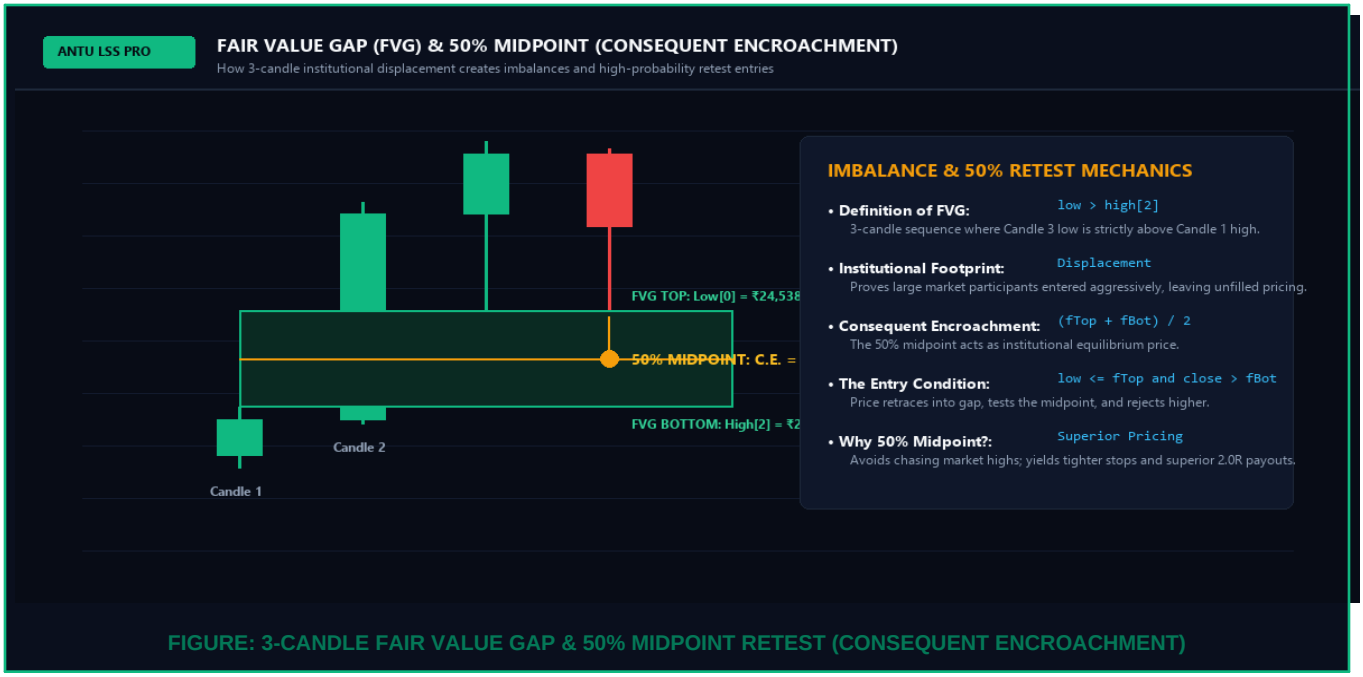


FIGURE: 3-CANDLE FAIR VALUE GAP & 50% MIDPOINT RETEST (CONSEQUENT ENCROACHMENT)

8. ANTU LSS PRO System Workflow

ANTU LSS PRO executes a disciplined 4-stage sequential state machine:

```
[STAGE 1: MONITORING]
Continuously calculate Pivot Highs (BSL) and Pivot Lows (SSL)
                  ↓
[STAGE 2: SWEEP DETECTION]
Price wicks through level and closes back inside (SSL Sweep or BSL Sweep)
Bias set to Bullish or Bearish; Setup State activated (within Session window)
                  ↓
[STAGE 3: DISPLACEMENT CONFIRMATION]
Wait for a subsequent 3-bar Fair Value Gap to form in reversal direction
Must occur within `expireBars` (default 30 bars)
                  ↓
[STAGE 4: RETEST ENTRY & RISK PROJECTION]
Price pulls back to retest the 50% midpoint of the post-sweep FVG
Entry line, Red SL Box (ATR-buffered), and Green TP Box (R:R ratio) plotted
TradingView alert dispatched immediately on candle close
```

9. The LONG Setup (Step-by-Step)

For a valid LONG signal to trigger, all of the following technical conditions must be satisfied in chronological order:

1. **Market Session Active:** The current bar time must fall within the configured session (default: 09:15–15:30 IST).
2. **No Active Open Position:** `hasOpen` must be `false` (the algorithm will not enter a new trade if an earlier trade is still unresolved).
3. **SSL Sweep Triggered:** Price wicks below an active Sell-Side Liquidity level and closes back above it (`low < lvl` and `close > lvl`).
4. **Setup State Initialized:**

• `lookLong` becomes `true`.

• Setup bar index is logged (`setupBarL = bar_index`).

- Stop-Loss baseline is established at `setupSLL = sweepLow - atr * atrMult`.
- If `showDebug` is enabled, an "SSL▲" label prints below the sweep candle.

5. **Bullish FVG Formation:** A bullish FVG (`low > high[2]`) forms *after* the sweep bar (`bFvgBar > setupBarL`).
6. **Retest of Midpoint:** Price retraces into the FVG zone such that `low <= fTop` and `close > fBot`.
7. **Signal Execution:** The trade executes at the FVG midpoint, plotting the green TP box, red SL box, and dashed entry line, while firing the LONG alert.

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10. The SHORT Setup (Step-by-Step)

For a valid SHORT signal to trigger, the inverse sequence is strictly verified:

1. **Market Session Active:** The bar must fall within active trading hours.
2. **No Active Open Position:** `hasOpen` must be `false`.
3. **BSL Sweep Triggered:** Price wicks above an active Buy-Side Liquidity level and closes back below it (`high > lvl` and `close < lvl`).
4. **Setup State Initialized:**

- `lookShort` becomes `true`.
- Setup bar index is logged (`setupBarS = bar_index`).
- Stop-Loss baseline is established at `setupSLS = sweepHigh + atr * atrMult`.
- If `showDebug` is enabled, a "BSL▼" label prints above the sweep candle.

5. **Bearish FVG Formation:** A bearish FVG (`high < low[2]`) forms *after* the sweep bar (`sFvgBar > setupBarS`).
6. **Retest of Midpoint:** Price retraces upward into the bearish FVG zone such that `high >= fBot` and `close < fTop`.
7. **Signal Execution:** The trade executes at the FVG midpoint, plotting the green TP box, red SL box, and dashed entry line, while firing the SHORT alert.

---

11. Entry Logic & Consequent Encroachment

Rather than entering at the extreme boundary of an imbalance—which exposes the trader to wider stop-losses—ANTU LSS PRO enters at the **50% midpoint** of the Fair Value Gap, known in institutional theory as **Consequent Encroachment (C.E.)**:

entryP = (fTop + fBot / 2)

Entering at the midpoint achieves three vital objectives:

1. Improves execution pricing by avoiding chasing momentum.
2. Reduces the total risk distance (slDist), allowing for tighter stop parameters.
3. Increases the nominal profit target payout while maintaining a 2.0 Risk-to-Reward ratio.

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12. Stop Loss Calculation (ATR Buffer)

Stop losses are never placed at arbitrary round numbers. ANTU LSS PRO anchors the stop loss beyond the extreme wick of the liquidity sweep, plus an added volatility buffer derived from the 14-period Average True Range (ATR):

- **For Long Trades:**

slP = sweepLow - (ATR_{14} x atrMult)

- **For Short Trades:**

slP = sweepHigh + (ATR_{14} x atrMult)

The default multiplier is **`0.5`**, meaning the stop loss is placed half an ATR beyond the sweep wick. This prevents market "micro-wicks" or spread fluctuations from stopping you out prematurely.

13. Take Profit Calculation (Fixed R:R Multiplier)

Take Profit is calculated strictly as a function of the risk distance:

1. Calculate Risk Distance (slDist):

• Long: slDist = entryP - slP

• Short: slDist = slP - entryP

2. Project Take Profit (tpP):

• Long: tpP = entryP + (slDist x rrRatio)

• Short: tpP = entryP - (slDist x rrRatio)

With the default **`rrRatio = 2.0`**, your potential profit is exactly twice your defined risk on every trade.



FIGURE: COMPLETE TRADE EXECUTION WITH ATR STOP LOSS AND 2.0R TARGET BOX

14. Risk-to-Reward Ratio (R:R) Dynamics

The mathematical advantage of a fixed 2.0 Risk-to-Reward ratio is fundamental to long-term trading longevity.

Win Rate %	Wins (at 2R)	Losses (at 1R)	Net Result (Units of Risk)	Outcome
35%	35 x 2 = +70R	65 x (-1) = -65R	**+5R**	Profitable
40%	40 x 2 = +80R	60 x (-1) = -60R	**+20R**	Highly Profitable
50%	50 x 2 = +100R	50 x (-1) = -50R	**+50R**	Exceptional

Because ANTU LSS PRO enforces a 2.0 R:R by default, you do not require an unrealistic 80% or 90% win rate to achieve positive expectancy. Even at a modest 40% win rate, a 2:1 R:R profile generates a net positive return over a large sample of disciplined trades.

15. Average True Range (ATR) Volatility Integration

ANTU LSS PRO continuously computes a 14-period Average True Range (`ta.atr(14)`).

Unlike fixed-point stop losses (e.g., "always use 20 points"), market volatility changes dynamically between quiet sessions and high-volatility events. By tying the stop buffer to `atr * atrMult`, the indicator automatically widens the protective buffer during volatile conditions and tightens it during calm conditions.

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16. Session Settings & Market Time Filters

The script incorporates a session filter using TradingView's native session syntax:

- **Default Value:** `"0915-1530"`
- **Default Timezone:** `"Asia/Kolkata"`

How the Session Filter Operates:

- When price is outside the defined session hours, `inSession` evaluates to `false`.
- If a setup was pending (`lookLong` or `lookShort`), it is immediately cancelled (`lookLong := false`, `lookShort := false`).
- No new setups can be triggered outside of the designated session.
- **For Non-Indian or 24/7 Markets (Forex, Crypto, US Equities):** Simply change the session string in the indicator settings to `"0000-2359"` or your local market exchange hours.

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17. Indicator Settings & Input Parameters

Input Parameter	Script Variable	Default	Valid Range	Function & Explanation
Swing Length	`swingLen`	`10`	3 – 50	Number of bars to the left and right required to confirm a pivot high/low. Lower values detect faster minor swings; higher values detect major swing levels.
Max Liq Levels	`maxLevels`	`5`	1 – 10	Maximum number of active un-swept SSL/BSL levels held in memory.
SL ATR Buffer	`atrMult`	`0.5`	0.1 – 5.0	Multiplier applied to 14 ATR added beyond the sweep wick for stop placement.
R:R Ratio	`rrRatio`	`2.0`	0.5 – 10.0	Risk-to-Reward ratio determining the Take Profit distance relative to SL.
Expire setup after N bars	`expireBars`	`30`	3 – 200	If an FVG retest does not occur within N bars following a sweep, the setup expires.
Stats: Last N Signals	`statsLen`	`20`	5 – 100	The rolling sample size of historical signals displayed in the top-right dashboard.
Show setup markers	`showDebug`	`true`	true / false	Toggles on/off visual "SSL▲" and "BSL▼" labels when sweeps occur.
Market Session	`sessionIndia`	`"0915-1530"`	String	Active trading hours (formatted HHMM-HHMM in Asia/Kolkata timezone).

Long Color	`longCol`	`#26a69a`	Color	Visual theme for long setup boxes and sweep labels.
Short Color	`shortCol`	`#ef5350`	Color	Visual theme for short setup boxes and sweep labels.
Box Width (bars)	`boxWidth`	`30`	5 – 100	Visual horizontal width in bars for newly plotted TP and SL boxes.

18. How to Read the Chart

Visual Overlays:

1. **"SSL▲" Label (Teal/Green):** Indicates that price swept a Sell-Side Liquidity level and closed back inside. The algorithm is now searching for a bullish FVG retest.
2. **"BSL▼" Label (Red):** Indicates that price swept a Buy-Side Liquidity level and closed back inside. The algorithm is now searching for a bearish FVG retest.
3. **Dashed White Line:** Marks the exact Entry price (FVG midpoint).
4. **Green Box (TP):** Visual Take-Profit target zone labeled with the exact price.
5. **Red Box (SL):** Visual Stop-Loss protection zone labeled with the exact price.
6. **"■ LONG" / "■ SHORT" Label:** Appears at the entry candle showing direction and exact price.

On-Chart Statistics Dashboard (Top Right):

- **Header:** "ANTU LSS PRO | Signals"
- **Bias:** Displays current directional bias ("Bullish ▲", "Bearish ▼", or "–").
- **Win Rate:** Displays historical win rate percentage over the sample window (colored green if >= 50\%, red if < 50\%).
- **Wins:** Total signals reaching TP within the sample window.
- **Losses:** Total signals hitting SL within the sample window.
- **Expired:** Setups that neither hit TP nor SL before reaching max hold duration (3 x expireBars).
- **Open:** Currently unresolved active trade signals.
- **Sample:** Number of closed signals evaluated out of `statsLen`.

19. Common Trading Mistakes to Avoid

1. **Trading FVGs Without a Preceding Sweep:** Never take an FVG in isolation. An FVG only holds strong predictive validity when institutional liquidity has been taken first.
2. **Chasing the Retest with Market Orders:** If price moves past the entry line before you execute, do not chase. Wait for a clean test of the 50% midpoint.
3. **Ignoring Session Filters on Non-Indian Assets:** If you trade Forex or US Equities, remember to adjust `sessionIndia` to your local trading session or `"0000-2359"`.
4. **Altering Stop Loss During a Trade:** The indicator calculates a volatility-buffered stop loss for a reason. Moving your stop loss wider during a trade destroys your risk profile.
5. **Over-Leveraging on Single Trades:** Never risk more than 1% to 2% of your total trading account equity on any individual trade signal.

20. Professional Risk Management Rules

Successful trading is 20% methodology and 80% risk management. Even the most robust technical setups experience losing streaks.

Recommended Position Sizing Formula:

Position Size (Quantity) = (Account Capital x Risk Percentage / Entry Price - Stop Loss Price)

Example Calculation:

- **Account Capital:** ■1,00,000
- **Risk Per Trade:** 1% (■1,000 max risk)
- **Entry Price:** ■2,500
- **Stop Loss Price:** ■2,480
- **Risk Distance (slDist):** ■20
- **Permitted Quantity:** (■1,000 / ■20) = 50 shares

If the trade hits stop loss, you lose exactly ■1,000 (1%). If it reaches the 2.0R take-profit at ■2,540, you gain ■2,000 (2%).

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21. Native TradingView Alerts Integration

ANTU LSS PRO features full alert integration. You can choose between:

Method A: Visual / Audio Popups (Standard Alertcondition)

- Create an alert on "ANTU LSS PRO Long", "ANTU LSS PRO Short", or "ANTU LSS PRO Any".
- Select **"Once Per Bar Close"**.
- Receive popups, mobile push alerts, and sound notifications when a candle confirms a setup.

Method B: Dynamic Structured Payload (Any alert() function call)

- Create an alert selecting **"Any alert() function call"** as the condition.
- The indicator automatically sends a dynamic JSON payload containing the exact calculated values:

```
{
  "action": "BUY",
  "signal": "LONG",
  "indicator": "ANTU LSS PRO",
  "ticker": "NIFTY",
  "exchange": "NSE",
  "interval": "5",
  "entry": 24520.50,
  "sl": 24480.25,
  "tp": 24601.00
}
```

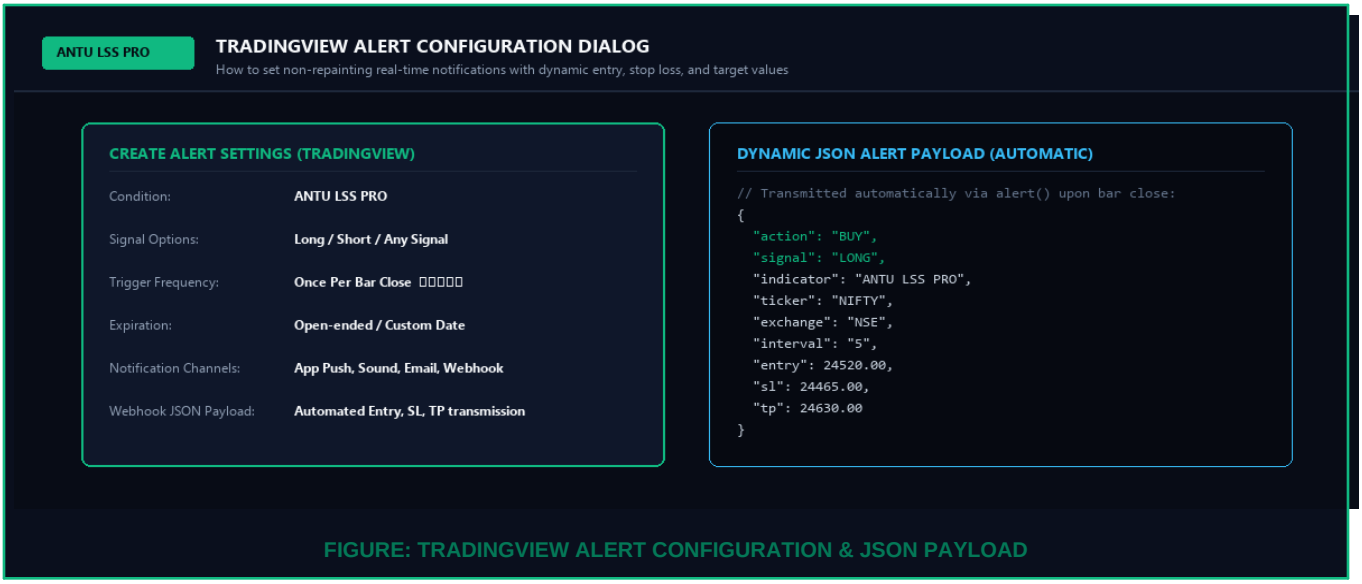


FIGURE: TRADINGVIEW ALERT CONFIGURATION & JSON PAYLOAD

22. Frequently Asked Questions (FAQ)

Q: Does ANTU LSS PRO repaint?

A: No. Historical signals and alert triggers calculate on confirmed candle data and execute on `alert.freq_once_per_bar_close`. Once a bar closes, the signal is permanently fixed.

Q: Can I use this on a free TradingView plan?

A: Yes. The script compiles and runs smoothly on the free TradingView Basic tier.

Q: Why did an "SSL▲" label appear, but no trade signal was triggered?

A: A sweep label marks the *potential* setup state. If price fails to produce a valid Fair Value Gap in the reversal direction within `expireBars`, or fails to retrace to the 50% midpoint, the setup expires safely without triggering a trade.

Q: How do I change the market session for crypto or forex?

A: Open the indicator settings (gear icon), navigate to the "Sessions" group, and change `"0915-1530"` to `"0000-2359"`.

23. Official Risk Disclaimer

ANTU LSS PRO is a technical analysis indicator and educational software tool. It does not provide financial, investment, legal, tax, or trading advice, and does not guarantee profits or positive returns. Financial trading carries substantial risk of monetary loss. Past performance, whether historical or simulated on a chart, does not guarantee future results. You are solely and completely responsible for all trading and investment decisions executed on your personal brokerage accounts.